
Smart Virtual Fitting Mirror System

Android Box + 65" Display + Tablet Controller

Project Overview

I want to build a professional Smart Virtual Fitting Mirror System for clothing stores.

The system will allow customers to virtually try on clothing and accessories in real time using a camera while standing in front of a 65-inch display.

The system must work completely offline without any internet connection.

The system consists of two Android applications that communicate over a local Wi-Fi network.

System Architecture

The Smart Mirror System consists of two connected applications.

1. Display Application (Android Box)

This application runs on an Android Box connected to:

- 65-inch Display
- USB Camera
- Local Database
- Product Images Storage

Responsibilities:

- Open the camera.
 - Detect the customer's body using MediaPipe Pose.
 - Overlay virtual clothing in real time.
 - Track body movement smoothly.
 - Display multiple products simultaneously.
 - Store all product information locally.
 - Receive commands from the Tablet Controller.
 - Work completely offline.
-

2. Tablet Controller Application

This application runs on an Android tablet.

Responsibilities:

- Browse products.
- Search products.
- Scan product barcode.

- Select clothing.
- Change product colors.
- Add or remove products.
- Manage the product catalog.
- Send commands to the Android Box over Wi-Fi.

The Tablet does not process camera images.

It only controls the Smart Mirror.

Communication

The Tablet Controller communicates with the Android Box using the local Wi-Fi network.

No internet connection is required.

Examples of commands:

- Select Product
- Change Color
- Remove Product
- Start Try-On
- Add New Product
- Update Product
- Delete Product

All commands are executed immediately on the Android Box.

Local Database

The local database is stored only on the Android Box.

The Tablet accesses and updates this database through the local Wi-Fi connection.

The system must support offline synchronization between both applications.

Admin Panel (Tablet)

The store owner should be able to manage products from the Tablet.

Each product contains:

- Product Name
- Barcode
- Product Category
- Available Colors
- Product Images

No inventory management is required.

No stock management is required.

No POS integration is required.

Product Categories

The system should initially support:

Clothing

- T-Shirts
- Shirts
- Jackets
- Hoodies
- Pants

Required Images:

- Front Image
 - Back Image
-

Shoes

Required Images:

- Front Left
 - Front Right
 - Side Left
 - Side Right
-

Glasses

Required Images:

- Front Image
 - Side Image
-

Hats

Required Images:

- Front Image
 - Back Image
-

The system should allow adding new product categories in the future without rebuilding the application.

Customer Experience

The customer stands in front of the Smart Mirror.

Using the Tablet Controller, the customer can:

- Browse products.
- Search by product name.
- Scan product barcode.
- Select a product.
- Change product color.
- Add multiple products.
- Remove products.
- Press "Try On".

The selected products immediately appear on the Smart Mirror.

Virtual Try-On

The Display Application must:

- Detect the customer's body.
 - Detect body landmarks using MediaPipe Pose.
 - Display clothing aligned with the body.
 - Track body movement smoothly.
 - Maintain stable alignment.
 - Run at approximately 30 FPS or higher.
 - Work completely offline.
-

Color Selection

If a product has multiple colors, the customer can instantly switch between them using the Tablet Controller.

The Smart Mirror updates immediately.

Multiple Products

The customer should be able to wear multiple products simultaneously.

Example:

- Shirt
- Pants
- Shoes
- Glasses
- Hat

All products should appear together correctly.

Barcode Scanner

The customer can scan a product barcode using the Tablet.

The corresponding product immediately appears on the Smart Mirror.

Smart Recommendations

After selecting a product, the system should recommend matching products from the local catalog.

Example:

After selecting a shirt:

- Matching Pants
 - Matching Shoes
 - Matching Glasses
 - Matching Hat
-

Offline Mode

The entire system must work without internet.

The Android Box stores:

- Database
- Product Images
- Application Data

The Tablet communicates only through the local Wi-Fi network.

Performance Requirements

The system should provide:

- Smooth body tracking.
 - Stable clothing alignment.
 - Approximately 30 FPS or higher.
 - Low latency between Tablet and Display.
 - Fast product switching.
 - Instant color switching.
-

Future Expansion

The system should be modular.

Future features may include:

- New clothing categories.
- Customer profiles.

- AI recommendations.
- Multiple Smart Mirrors.
- Cloud synchronization.
- Inventory integration.
- POS integration.

These features should be easy to add later.

Source Code

The developer must provide:

- Complete Android Studio source code.
 - Android Box project.
 - Tablet Controller project.
 - APK for Android Box.
 - APK for Tablet.
 - Local database.
 - Documentation.
 - Installation guide.
 - Communication documentation.
 - Network configuration guide.
-

Ownership

The client receives:

- Full commercial ownership.
- Full source code.
- All project files.
- No encrypted code.
- No locked libraries.
- No usage restrictions.

The developer may not reuse or resell the source code without permission.

Project Summary

The Smart Virtual Fitting Mirror System consists of:

- Android Box
- USB Camera
- 65-inch Display
- Tablet Controller

- Offline Local Database
- MediaPipe Pose Tracking
- Real-Time Virtual Try-On

Customers browse products using the Tablet Controller while standing in front of the Smart Mirror.

The Tablet sends commands through the local Wi-Fi network.

The Android Box instantly displays the selected clothing on the customer's body in real time.

The entire system works completely offline and includes complete source code with full commercial ownership.
